0013

TEST REPORT

Lucideon Reference: N26458 (QT-78966/4/KNA)/Ref. 6

Project Title: Shear Testing to BS EN 1052-3:2002 of L10 Low Carbon Brick

Client: Earth4Earth Technology Ltd
85 Great Portland Street
London
W1W 7LT

For the Attention of: Ms Anna Simmons

Author(s): Mr Kaleem Zar

Report Date: 26 May, 2026

Purchase Order No.: PO-0003

Opinions and interpretations expressed herein are outside the scope of UKAS Accreditation.

Mr Justin Fryer
Testing Team
Reviewer

K. Zar

Mr Kaleem Zar
Testing Team
Project Manager

AMRICC INSIDE

Lucideon Limited
Queens Road, Penkhull
Stoke-on-Trent
Staffordshire, ST4 7LQ, UK

T +44 (0)1785 331874
query@lucideon.com
www.lucideon.com
Reg. England 1960455

CONTENTS

1 INTRODUCTION

Earth4Earth Technology Ltd required a programme of testing to determine the shear strength of masonry samples comprising of L10 Low Carbon bricks in accordance with BS EN 1052-3:2002 Methods of Test for Masonry – Part 3: Determination of Initial Shear Strength.

Testing was carried out on 7 May 2026 at Lucideon's Structures Laboratory at Queens Road, Penkhull, Stoke-on-Trent, ST4 7LQ.

2 SAMPLES RECEIVED

Solid bricks referenced as L10 Low Carbon Bick were received from the client.

The bricks were nominally 215 mm x 102.5 mm x 65 mm (length x width x height).

The bricks were solid with a nominal compressive strength of 50.9 N/mm².

3 METHOD OF CONSTRUCTION

Nine triplet samples were constructed by a bricklayer from Lucideon with M4 mortar applied at each course.

The Mortar had a compressive strength of 4.16 N/mm^2.

Following construction, the samples were pre-compressed by a uniform mass to give a vertical stress of 2.07 x 10^{-3} N/mm^2.

The samples were close covered with polythene sheets for 28 days at ambient laboratory temperatures.

The samples were constructed on 9 April 2026.

4 METHOD OF TEST

The sample was placed in the shear test rig, such that, the faces of the masonry units, where the shear load is applied, were plane and perpendicular to the direction of the shear load.

The rate of applied load was in between 0.1 N/mm^2 per minute and 0.4 N/mm^2 per minute, which was within the limits given in BS EN 1052-3:2002.

Three tests at each of the precompressions of 0.2, 0.6 and 1.0 N/mm^2 were carried out following Procedure A in BS EN 1052-3:2002.

The testing was carried out under ambient laboratory temperature and humidity.

5 RESULTS

Individual shear results are shown in the Tables Section.

A plot of shear strength versus precompression can be seen for every test in the Chart Section.

The failure mode is shown in Plate 1.

6 SUMMARY

The summary of results calculated based on data recorded during testing can be seen in the Tables Section.

TABLES

Table 1 – Mortar Properties

*Compressive Strength to EN 1015-11 (N/mm^2) *	*Air Content to EN 1015-7 (%)	*Flow Value to BS 4551
4.16	5.1	174

**These tests are not on Lucideon's UKAS Schedule.*

Table 2 – Shear Results for L10 Low Carbon Bricks Tested in Accordance with BS EN 1052-3:2002

Sample No.	$F_{i,max}$ Maximum Shear Load (N)	Length (mm)	Width (mm)	f_{pi} Precompressive Stress of Individual Sample (N/mm^2)	f_{voi} Shear Strength of Individual Sample (N/mm^2)	Failure Mode
1	12500	215	100	0.2	0.20	A1 in accordance with BS EN 1052-3
2	10910	215	100		0.22	
3	10250	215	102		0.21	
4	16860	215	102	0.6	0.47	
5	17800	215	102		0.44	
6	20850	215	102		0.46	
7	28890	215	103	1.0	0.65	
8	27560	215	100		0.65	
9	27980	215	102		0.61	

Table 3 − The Summary of Testing Results for L10 Low Carbon Bricks Tested in Accordance with BS EN 1052-3:2002

Name of the Characteristic	Value
Mean initial shear strength was determined	0.15 N/mm^2
Internal angle of friction	25.66°
Characteristic initial shear strength (Procedure A)	0.12 N/mm^2
Characteristic angle of internal friction	21.02°

NOTE: The results given in this report apply only to the samples that have been tested.

END OF REPORT

PLATE

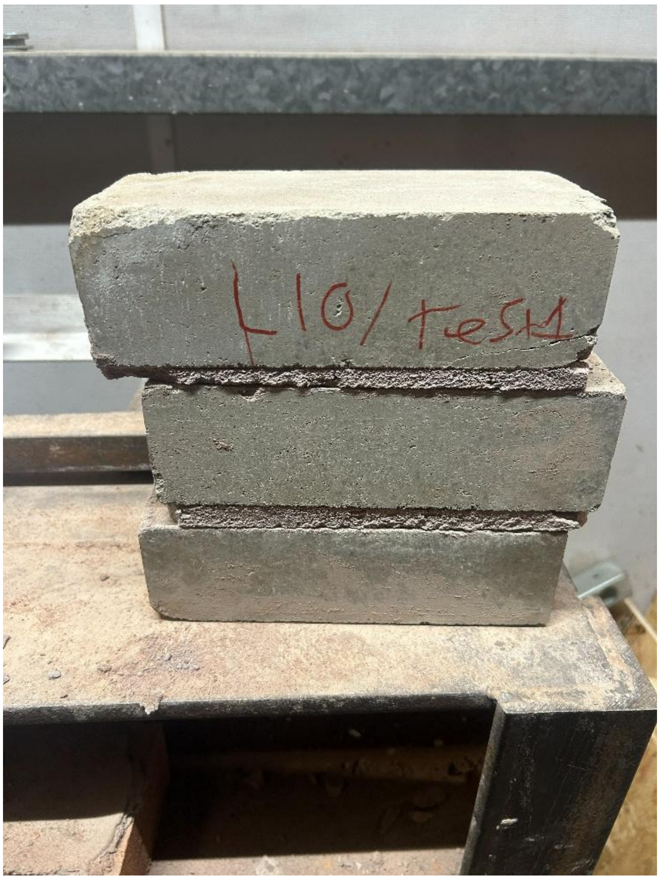

Plate 1 – Typical Shear Test Failure

Chart 1 - Shear Strength vs Pre-Compression Chart for L10 Low Carbon Bricks